GATE CS — Theory of Computation

Topic-wise past questions (2016–2026) + new practice questions with solutions

Built from the papers in pAPER/ of the GATE2027 repo. Topics are ordered by total marks (most important first).

Topic	Past questions	Total marks	New practice Qs
CFG / PDA / CFL	33	54	2
Regular languages, DFA/NFA, regex	27	40	2
Turing machines, decidability	13	21	2
Total	**73**	**115**	**6**

How to read this

- Each topic has **A. Past-paper questions** (newest first; tagged year · set · question number · marks) and **B. New practice questions** with short solutions.
- Past questions have **no answer key** here; check them against the official answer keys.
- Question text was extracted from the PDFs. Code, tables, symbols and figures may be flattened or lost. Entries marked [OCR] come from scanned papers and may contain errors; open the original in pAPER/<year>/ for the exact wording.
- New practice questions are original, in GATE style; numerical and program-output answers were verified.

CFG / PDA / CFL

CFL vs non-CFL, PDA/DPDA, grammar properties, closure. · 33 questions, 54 marks over 15 papers (≈3.6 marks/paper)

A. Past-paper questions (33)

2026 CS1 · Q.25 · 1 mark
Consider the following grammar where S is the start symbol, and a and b are terminal symbols. S → aSbS | bS | ε Which of the following statements is/are true?
(A) The grammar is ambiguous
(B) The string abb has two distinct derivations in this grammar
(C) The string abab has only one rightmost derivation
(D) The language generated by the grammar is undecidable

2026 CS1 · Q.52 · 2 marks
Consider the following context-free grammar G. S → abaABAbba A → aaBBAb | bBabaa B → aBb | ab In the above grammar, S is the start symbol, a and b are terminal symbols, and A and B are non-terminal symbols. Let L(G) be the language generated by the grammar G. For a string s ∈ L(G), let n1 (s) be the number of a's in s and n2 (s) be the number of b's in s. Which of the following statements is/are true?
(A) There is a string s ∈ L(G) such that n1 (s) < n2 (s)
(B) For every string s ∈ L(G), n1 (s) ≥ n2 (s)
(C) There is a string s ∈ L(G) such that n1 (s) > 2n2 (s)
(D) For every string s ∈ L(G), n1 (s) ≤ 2n2 (s)

2026 CS2 · Q.29 · 1 mark
Which of the following grammars is/are ambiguous?
(A) S → aSb | ε
(B) E → E + E | E ∗ E | id
(C) S → aS | Sa | ε
(D) S → aS | ε

2026 CS2 · Q.48 · 2 marks
Let Σ = {a, b, c, d} and let L = {ai b j c k d l | i, j, k, l ≥ 0}. Which of the following constraints ensure(s) that the language L is context-free?
(A) i+k = j + l
(B) i = k and j = l
(C) i = l and j = k
(D) i+j = k + l

2025 CS1 · Q.19 · 1 mark
Consider the following context-free grammar G, where S, A, and B are the variables (non-terminals), a and b are the terminal symbols, S is the start variable, and the rules of G are described as: S → aaB | Abb A → a | aA B → b | bB Which ONE of the languages L(G) is accepted by G?
(A) L(G) = {a2 b n | n ≥ 1} ∪ {an b 2 | n ≥ 1}
(B) L(G) = {an b 2n | n ≥ 1} ∪ {a2n b n | n ≥ 1}
(C) L(G) = {an b n | n ≥ 1}
(D) L(G) = {a2n b 2n | n ≥ 1}

2025 CS1 · Q.44 · 2 marks
Consider the following two languages over the alphabet {a, b}: L1 = { αβα | α ∈ {a, b}+ AND β ∈ {a, b}+ } L2 = { αβα | α ∈ {a}+ AND β ∈ {a, b}+ } Which ONE of the following statements is CORRECT?
(A) Both L1 and L2 are regular languages.
(B) L1 is a regular language but L2 is not a regular language.
(C) L1 is not a regular language but L2 is a regular language.
(D) Neither L1 nor L2 is a regular language.

2025 CS1 · Q.45 · 2 marks
Consider the following two languages over the alphabet {a, b, c}, where m and n are natural numbers. L1 = {am b m c m+n | m, n ≥ 1} L2 = {am b n c m+n | m, n ≥ 1} Which ONE of the following statements is CORRECT?
(A) Both L1 and L2 are context-free languages.
(B) L1 is a context-free language but L2 is not a context-free language.
(C) L1 is not a context-free language but L2 is a context-free language.
(D) Neither L1 nor L2 are context-free languages.

2025 CS2 · Q.24 · 1 mark
Which ONE of the following languages is accepted by a deterministic pushdown automaton?
(A) Any regular language.
(B) Any context-free language.
(C) Any language accepted by a non-deterministic pushdown automaton.
(D) Any decidable language.

2025 CS2 · Q.25 · 1 mark
Let G1 , G2 be Context Free Grammars (CFGs) and R be a regular expression. For a grammar G, let L(G) denote the language generated by G. Which ONE among the following questions is decidable?
(A) Is L(G1) = L(G2)?
(B) Is L(G1) ∩ L(G2) = ∅?
(C) Is L(G1) = L(R)?
(D) Is L(G1) = ∅?

2025 CS2 · Q.30 · 1 mark
Consider the two lists List I and List II given below: List I List II (i) Context free languages (a) Closed under union (ii) Recursive languages (b) Not closed under complementation (iii) Regular languages (c) Closed under intersection For matching of items in List I with those in List II, which of the following option(s) is/are CORRECT?
(A) (i) – (a), (ii) – (b), and (iii) – (c)
(B) (i) – (b), (ii) – (a), and (iii) – (c)
(C) (i) – (b), (ii) – (c), and (iii) – (a)
(D) (i) – (a), (ii) – (c), and (iii) – (b)

2025 CS2 · Q.52 · 2 marks
Let Σ = {a, b, c}. For x ∈ Σ ∗ , and α ∈ Σ, let #α (x) denote the number of occurrences of α in x. Which one or more of the following option(s) define(s) regular language(s)?
(A) {am b n | m, n ≥ 0}
(B) {a, b}∗ ∩ {am b n c m−n | m ≥ n ≥ 0}
(C) {w | w ∈ {a, b}∗ , #a (w) ≡ 2 (mod 7), and #b (w) ≡ 3 (mod 9)}
(D) {w | w ∈ {a, b}∗ , #a (w) ≡ 2 (mod 7), and #a (w) = #b (w)}

2024 CS1 · Q.59 · 2 marks
Let G = (V, Σ, S, P) be a context-free grammar in Chomsky Normal Form with Σ = {a, b, c} and V containing 10 variable symbols including the start symbol S. The string w = a30 b 30 c 30 is derivable from S. The number of steps (application of rules) in the derivation S →∗ w is _____

2024 CS2 · Q.52 · 2 marks
Consider a context-free grammar G with the following 3 rules. S → aS, S → aSbS, S → c Let w ∈ L(G). Let na (w), nb (w), nc (w)denote the number of times a, b, c occur in w, respectively. Which of the following statements is/are TRUE?
(A) na (w) > nb (w)
(B) na (w) > nc (w) − 2
(C) nc (w) = nb (w) + 1
(D) nc (w) = nb (w) ∗ 2

2023 · Q.24 · 1 mark
Which of the following statements is/are CORRECT?
(A) The intersection of two regular languages is regular.
(B) The intersection of two context-free languages is context-free.
(C) The intersection of two recursive languages is recursive.
(D) The intersection of two recursively enumerable languages is recursively enumerable.
CS

2023 · Q.39 · 2 marks
Consider the context-free grammar G below S → aSb | X X → aX | Xb | a | b , where S and X are non-terminals, and a and b are terminal symbols. The starting non-terminal is S. Which one of the following statements is CORRECT?
(A) The language generated by G is (a + b)∗
(B) The language generated by G is a∗ (a + b)b∗
(C) The language generated by G is a∗ b∗ (a + b)
(D) The language generated by G is not a regular language CS

2023 · Q.40 · 2 marks
Consider the pushdown automaton (PDA) P below, which runs on the input alpha- bet {a, b}, has stack alphabet {⊥, A}, and has three states {s, p, q}, with s being the start state. A transition from state u to state v, labelled c/X/γ, where c is an input symbol or , X is a stack symbol, and γ is a string of stack symbols, represents the fact that in state u, the PDA can read c from the input, with X on the top of its stack, pop X from the stack, push in the string γ on the stack, and go to state v. In the initial configuration, the stack has only the symbol ⊥ in it. The PDA accepts by empty stack. a/⊥/A⊥ a/A/AA b/A/ b/A/ s p /A/ /A/ q /A/ /⊥/ Which one of the following options correctly describes the language accepted by P ?
(A) {am bn | 1 ≤ m and n < m}
(B) {am bn | 0 ≤ n ≤ m}
(C) {am bn | 0 ≤ m and 0 ≤ n}
(D) {am | 0 ≤ m} ∪ {bn | 0 ≤ n} CS

2022 · Q.47 · 2 marks
Consider the following languages: L1 {a n wa n | w {a, b}*} L2 {wxwR | w, x {a, b}* , | w |,| x | 0} Note that w R is the reversal of the string w. Which of the following is/are TRUE?
(A) L1 and L2 are regular.
(B) L1 and L2 are context-free.
(C) L1 is regular and L2 is context-free.
(D) L1 and L2 are context-free but not regular.

2022 · Q.48 · 2 marks
Consider the following languages: L1 {ww | w {a, b}*} L2 {a n bn c m | m, n 0} L3 {a mbn cn | m, n 0} Which of the following statements is/are FALSE?
(A) L1 is not context-free but L2 and L3 are deterministic context-free.
(B) Neither L1 nor L2 is context-free.
(C) L2 , L3 and L2 L3 all are context-free.
(D) Neither L1 nor its complement is context-free.

2021 · Q.1 · 1 mark [OCR]
Suppose that L, is a regular language and Ly» is a context-free language. Which one of the following languages is NOT necessarily context-free?
(A) | Lin Le
(B) | Ly - Le
(C) | Li—L2
(D) | Li UL,

2021 · Q.51 · 2 marks [OCR]
In a pushdown automaton P = (Q,,T,4,qo, F), a transition of the form, Oo a,X +Y ® where p,q € Q, a € UU {e}, and X,Y ET u {e}, represents (4, Y) € d(p,a, X). Consider the following pushdown automaton over the input alphabet © = {a,b} and stack alphabet = {#, A}. ae+A bA+e 66€3# a ee e€ 6 AsA start —(40 n .) The number of strings of length 100 accepted by the above pushdown automaton is

2020 · Q.10 · 1 mark [OCR]
Consider the language L= {a"|n>0}U {a"b"|n > 0} and the following statements. I. L is deterministic context-free. Il. L is context-free but not deterministic context-free. III. Z is not LL
(A) for any k. Which of the above statements is/are TRUE?
(A) lonly (8) Il only (©) land III only
(D) ILL only

2020 · Q.32 · 2 marks [OCR]
Consider the following languages. L, = {wxyx | w,x,y € (0+ 1)*} Lz = {xy |x y € (a+ by', |x] = ly x # y} Which one of the following is TRUE?
(A) L, is regular and L» is context-free. (8) L, is context-free but not regular and L, is context-free. () Neither L, nor L» is context-free. (0) L, is context-free but Lz is not context-free.

2019 · Q.31 · 2 marks [OCR]
Which one of the following languages over 2 = {a, b} is NOT context-free?
(A) {ww® |w € {a, b}}
(B) {wa"b"w |w € {a, b}",n = 0}
(C) {wa"w® b" |w € {a, b}*,n = 0}
(D) {abi | i € {n,3n, 5n},n = 0} cs TAS GATE 2019 Computer Science and Information Technology Set-2

2018 · Q.35 · 2 marks
Consider the following languages: I. {am b n c p d q | m + p = n + q, where m, n, p, q ≥ 0} II. {am b n c p d q | m = n and p = q, where m, n, p, q ≥ 0} III. {am b n c p d q | m = n = p and p ≠ q, where m, n, p, q ≥ 0} IV. {am b n c p d q | mn = p + q, where m, n, p, q ≥ 0} Which of the languages above are context-free?
(A) I and IV only
(B) I and II only
(C) II and III only
(D) II and IV only

2017 CS1 · Q.10 · 1 mark [OCR]
Consider the following context-free grammar over the alphabet ¥, = {a, b,c} with S as the start symbol: S — abScT | abcT T— bT|b Which one of the following represents the language generated by the above grammar?
(A) {(ab)"(cb)" |n = 1}
(B) {(ab)"cbTM cbhTM ... chTM | n,m4,m3,...,M, 21}
(C) {(ab)"(cbTM)" | m,n = 1}
(D) {(ab)"(cb")TM |m,n = 1}

2017 CS1 · Q.34 · 2 marks [OCR]
If G is a grammar with productions S = SaS | aSb | bSa|SS|e€ where S is the start variable. then which one of the following strings is not generated by G?
(A) abab
(B) aaab
(C) abbaa
(D) babba =====PAGE 16=====

2017 CS1 · Q.37 · 2 marks [OCR]
Consider the context-free grammars over the alphabet {a.b.c} given below. § and T are non- terminals. G,:S > aSb|T, T > cT\e G2:S > bSalT, T > cTle The language L(G,) N L(G) is
(A) Finite.
(B) Not finite but regular.
(C) Context-Free but not regular.
(D) Recursive but not context-free.

2017 CS1 · Q.38 · 2 marks [OCR]
Consider the following languages over the alphabet ¥ = fa, b, c}. Let Li= {aTMb"cTM|m,n = O}andL2= {aTMb"cTM | m,n = 0}. Which of the following are context-free languages? I. Li UL IL. Lin L2
(A) I only
(B) II only
(C) I and II
(D) Neither I nor II

2017 CS2 · Q.4 · 1 mark [OCR]
Let L,, Lz be any two context-free languages and R be any regular language. Then which of the following is/are CORRECT? I. L, UL, is context-free. II. L, is context-free. Ill. L, — R is context-free. IV. L, N Lz is context-free.
(A) L ILandIV only
(B) Iand III only
(C) ILand IV only
(D) Ionly

2017 CS2 · Q.16 · 1 mark [OCR]
Identify the language generated by the following grammar, where S is the start variable. S > XY X->aXla Y > aYble
(A) {aTMb" |m=n, n>0}
(B) {aTMb" |m=>=n, n=0}
(C) {aTMbTM |m>n, n=0}
(D) fab" |m>n, n>0}

2017 CS2 · Q.40 · 2 marks [OCR]
Consider the following languages. L, = {a? | p is a prime number} Lz = {a"bTMc?TM |n = 0,m = 0} E SfarbteeTM |= 0} Ly, = {a"b" |n > 1} Which of the following are CORRECT? I. L, is context-free but not regular. II. L, is not context-free. III. L3 is not context-free but recursive. IV. Lg is deterministic context-free. (A), IlandIV only
(B) ILand III only
(C) Tand IV only
(D) II and IV only =====PAGE 16=====

2016 · Q.42 · 2 marks
Consider the following context-free grammars: G1 : S → aS|B, B → b|bB G2 : S → aA|bB, A → aA|B|ε, B → bB|ε Which one of the following pairs of languages is generated by G1 and G2 , respectively?
(A) {am bn |m > 0 or n > 0} and {am bn |m > 0 and n > 0}
(B) {am bn |m > 0 and n > 0} and {am bn |m > 0 or n ≥ 0}
(C) {am bn |m ≥ 0 or n > 0} and {am bn |m > 0 and n > 0}
(D) {am bn |m ≥ 0 and n > 0} and {am bn |m > 0 or n > 0}

2016 · Q.43 · 2 marks
Consider the transition diagram of a PDA given below with input alphabet Σ = {a, b} and stack alphabet Γ = {X, Z}. Z is the initial stack symbol. Let L denote the language accepted by the PDA. a, X/XX a, Z/XZ b, X/ε b, X/ε ε, Z/Z Which one of the following is TRUE?
(A) L = {an bn |n ≥ 0} and is not accepted by any finite automata
(B) L = {an |n ≥ 0} ∪ {an bn |n ≥ 0} and is not accepted by any deterministic PDA
(C) L is not accepted by any Turing machine that halts on every input
(D) L = {an |n ≥ 0} ∪ {an bn |n ≥ 0} and is deterministic context-free

B. New practice questions with solutions

1. (1 mark · MCQ) Let L = { $a^i b^j c^k$ | i = j or j = k }. Which statement is TRUE?

A. L is regular.
B. L is context-free but not deterministic context-free.
C. L is deterministic context-free.
D. L is not context-free.

(B). L is the union of two CFLs ($a^n b^n c^*$ and $a^* b^n c^n$). A PDA must guess which condition to check, and a deterministic PDA cannot make that choice.

2. (2 marks · MSQ) Let L_1 = { a^n b^n c^m | n, m ≥ 0 } and L_2 = { a^n b^m c^m | n, m ≥ 0 }. Which statements are TRUE?

A. $L_1 \cap L_2$ is not context-free.
B. $L_1 \cup L_2$ is context-free.
C. The complement of $L_1 \cap L_2$ is context-free.
D. Context-free languages are closed under intersection.

(A), (B), (C). $L_1 \cap L_2$ = { a^n b^n c^n }, which is not context-free. The union of two CFLs is a CFL. The complement of { $a^n b^n c^n$ } is a union of CFLs (strings of the wrong shape, or with unequal counts), so it is context-free. Statement (D) is false, as (A) shows.

Regular languages, DFA/NFA, regex

Minimal DFA states, regex <-> DFA, closure properties, pumping lemma. · 27 questions, 40 marks over 15 papers (≈2.7 marks/paper)

A. Past-paper questions (27)

2026 CS1 · Q.26 · 1 mark
Let M be a nondeterministic finite automaton (NFA) with 6 states over a finite alphabet. Which of the following options CANNOT be the number of states in the minimal deterministic finite automaton (DFA) that is equivalent to M ?
(A) 32
(B) 65
(C) 1
(D) 128

2026 CS1 · Q.51 · 2 marks
Let L1 and L2 be two languages over a finite alphabet, such that L1 ∩ L2 and L2 are regular languages. Which of the following statements is/are always true?
(A) L1 is regular
(B) L1 ∪ L2 is regular
(C) L2 is context-free
(D) L1 is context-free

2026 CS2 · Q.47 · 2 marks
Consider the following two finite automata D1 and D2 . Which of the following statements is/are true?
(A) L(D1) = L(D2)
(B) L(D1) is a proper subset of L(D2)
(C) L(D1) ∩ L(D2) = { ε} ∗
(D) (L(D1) ∪ L(D2)) consists of all strings in {0,1}∗ whose length is divisible by 3

2025 CS1 · Q.28 · 1 mark
A regular language L is accepted by a non-deterministic finite automaton (NFA) with n states. Which of the following statement(s) is/are FALSE?
(A) L may have an accepting NFA with < n states.
(B) L may have an accepting DFA with < n states.
(C) There exists a DFA with ≤ 2n states that accepts L.
(D) Every DFA that accepts L has > 2n states.

2025 CS1 · Q.50 · 2 marks
Consider the following deterministic finite automaton (DFA) defined over the alphabet, Σ = {a, b}. Identify which of the following language(s) is/are accepted by the given DFA.
(A) The set of all strings containing an even number of b’s.
(B) The set of all strings containing the pattern bab.
(C) The set of all strings ending with the pattern bab.
(D) The set of all strings not containing the pattern aba.

2025 CS2 · Q.60 · 2 marks
Let Σ = {1,2,3,4}. For x ∈ Σ ∗ , let prod(x) be the product of symbols in x modulo 7. We take prod(ε) = 1, where ε is the null string. For example, prod(124) = (1 × 2 × 4) mod 7 = 1. Define L = {x ∈ Σ ∗ | prod(x) = 2}. The number of states in a minimum state DFA for L is _____. (Answer in integer)

2024 CS1 · Q.23 · 1 mark
Let L1 , L2 be two regular languages and L3 a language which is not regular. Which of the following statements is/are always TRUE?
(A) L1 = L2 if and only if L1 $\overset{\equiv}{\cap}$ L2 = φ
(B) L1 ∪ L3 is not regular
(C) $\overset{\equiv}{}$ L3 is not regular
(D) $\overset{\equiv}{}$ L1 $\overset{\equiv}{\cup}$ L2 is regular

2024 CS1 · Q.50 · 2 marks
Consider the 5-state DFA M accepting the language L(M) ⊂ (0 + 1)∗ shown below. For any string w ∈ (0 + 1)∗ let n0 (w) be the number of 0′ s in w and n1 (w) be the number of 1′s in w . Which of the following statements is/are FALSE?
(A) States 2 and 4 are distinguishable in M
(B) States 3 and 4 are distinguishable in M
(C) States 2 and 5 are distinguishable in M
(D) Any string w with n0 (w) = n1 (w) is in L(M)

2024 CS1 · Q.61 · 2 marks
Consider the following two regular expressions over the alphabet {0,1}: r = 0∗ + 1∗ s = 01∗ + 10∗ The total number of strings of length less than or equal to 5, which are neither in r nor in s, is _____

2024 CS2 · Q.22 · 1 mark
Which one of the following regular expressions is equivalent to the language accepted by the DFA given below? 0 1 0 1
(A) 0∗ 1(0 + 10∗ 1)∗
(B) 0∗ (10∗ 11)∗ 0∗
(C) 0∗ 1(010∗ 1)∗ 0∗
(D) 0(1 + 0∗ 10∗ 1)∗ 0∗

2024 CS2 · Q.41 · 2 marks
Let M be the 5-state NFA with ε-transitions shown in the diagram below. Which one of the following regular expressions represents the language accepted by M ?
(A) (00)∗ + 1(11)∗
(B) 0∗ + (1 + 0(00)∗)(11)∗
(C) (00)∗ + (1 + (00)∗)(11)∗
(D) 0+ + 1(11)∗ + 0(11)∗

2024 CS2 · Q.62 · 2 marks
and L2 = {w ∈ (a + b)∗ ||w| ≤ 4}, where |w| denotes the length of string w . The number of strings in L2 which are also in L1 is _____

2023 · Q.14 · 1 mark
Consider the Deterministic Finite-state Automaton (DFA) A shown below. The DFA runs on the alphabet {0, 1}, and has the set of states {s, p, q, r}, with s being the start state and p being the only final state. 0 1 1 s p q 1 0 0 r 0,1 Which one of the following regular expressions correctly describes the language accepted by A?
(A) 1(0∗ 11)∗
(B) 0(0 + 1)∗
(C) 1(0 + 11)∗
(D) 1(110∗)∗ CS

2023 · Q.63 · 2 marks
Consider the language L over the alphabet {0, 1}, given below: L = {w ∈ {0, 1}∗ | w does not contain three or more consecutive 1's}. The minimum number of states in a Deterministic Finite-State Automaton (DFA) for L is . CS

2022 · Q.12 · 1 mark
Which one of the following regular expressions correctly represents the language of the finite automaton given below?
(A) ab*bab* ba* aba*
(B) (ab*b)* ab* (ba*a)* ba*
(C) (ab*b ba*a)* (a* b*)
(D) (ba*a ab*b)* (ab* ba*)

2021 · Q.38 · 2 marks [OCR]
Consider the following language. L= {w € {0,1}° | w ends with the substring 011} Which one of the following deterministic finite automata accepts L? 0 0 1 1 tart & 29 8 yy 1. start > " 1 0
(A) 01 1 0 0 start & 0 ——o- 1 &) 1 0 0 1 start _& 0 &= oP No 1 1 ae ry start _& 0 — 0 (B (C)
(D) CS, Set-1 - Copyright © GATE 2021

2020 · Q.7 · 1 mark [OCR]
Which one of the following regular expressions represents the set of all binary strings with an odd number of 1°s?
(A) ((O + 1)*1(0 + 1)*1)*10* () (0°10*10")*0"1 (4) 10*(0°10*10")* (0) (0°10*10")*10°

2020 · Q.8 · 1 mark [OCR]
Consider the following statements. I. IfL, UL, is regular, then both L; and Lz must be regular. Il. The class of regular languages is closed under infinite union. Which of the above statements is/are TRUE?
(A) only (8) Il only (©) Both I and II (0) Neither I nor II

2020 · Q.51 · 2 marks [OCR]
Consider the following language. L = {x € {a, b}*| number of a's in x is divisible by 2 but not divisible by 3} The minimum number of states in a DFA that accepts L is

2019 · Q.15 · 1 mark [OCR]
For X= {a,b}, let us consider the regular language L = {x |x =a’ orx = k > 0}. Which one of the following can be a pumping length (the constant guaranteed by the pumping lemma) for L ? (A)3 (B)5 (9
(D) 24

2018 · Q.6 · 1 mark
Let N be an NFA with n states. Let k be the number of states of a minimal DFA which is equivalent to N. Which one of the following is necessarily true?
(A) k ≥ 2n
(B) k ≥ n
(C) k ≤ n2
(D) k ≤ 2n

2018 · Q.52 · 2 marks
Given a language L, define Li as follows: L0 = {ε} Li = Li−1 · L for all i > 0 The order of a language L is defined as the smallest k such that Lk = Lk+1 . Consider the language L1 (over alphabet 0) accepted by the following automaton. 0 0 0 The order of L1 is _____. CS 18/20 GATE 2018

2017 CS1 · Q.22 · 1 mark [OCR]
. . a * Consider the language Z given by the regular expression (4+) b(a+b) over the alphabet {a,b}. The smallest number of states needed in a deterministic finite-state automaton (DFA) accepting L is =====PAGE 12=====

2017 CS2 · Q.25 · 1 mark [OCR]
The minimum possible number of states of a deterministic finite automaton that accepts the regular language L = {wiaw2 | Wi, we € {a, b}*, | wi] = 2, | w2| = 3} is

2017 CS2 · Q.39 · 2 marks [OCR]
Let & denote the transition function and 6 denote the extended transition function of the e-NFA whose transition table is given below: 5 € a b —~ do {q2} {di} {qo} di tq2} {qa} {ds} q2 {do} 0) @ q3 0 iy {q2} Then &(qr. aba) is (A)o
(B) {do, 41,43}
(C) {4o, 41,42}
(D) {40.42.43}

2016 · Q.16 · 1 mark
Which of the following languages is generated by the given grammar? S −→ aS | bS | ε
(A) {an bm | n, m ≥ 0}
(B) {w ∈ {a, b}∗ | w has equal number of a’s and b’s }
(C) {an | n ≥ 0} ∪ {bn | n ≥ 0} ∪ {an bn | n ≥ 0}
(D) {a, b}∗

2016 · Q.18 · 1 mark

Which one of the following regular expressions represents the language: the set of all binary strings having two consecutive 0s and two consecutive 1s?

(A) (0 + 1)∗ 0011(0 + 1)∗ + (0 + 1)∗ 1100(0 + 1)∗

(B) (0 + 1)∗ (00(0 + 1)∗ 11 + 11(0 + 1)∗ 00)(0 + 1)∗

(C) (0 + 1)∗ 00(0 + 1)∗ + (0 + 1)∗ 11(0 + 1)∗

(D) 00(0 + 1)∗ 11 + 11(0 + 1)∗ 00

B. New practice questions with solutions

1. (1 mark · NAT) Over {a, b}, let L be the set of strings whose third symbol from the right is a. The number of states in the minimal complete DFA for L is _____.

8. A DFA must remember the last 3 symbols, and all $2^3 = 8$ histories are pairwise distinguishable.

2. (2 marks · MSQ) Which of the following languages are regular?

A. $\{ a^n b^m \mid n, m \geq 0 \text{ and } n + m \text{ is even} \}$

B. $\{ a^n b^n \mid n \geq 0 \}$

C. $\{ w \in \{a, b\}^* \mid \#_a(w) \equiv \#_b(w) \pmod 3 \}$

D. $\{ a^n \mid n \text{ is a perfect square} \}$

(A), (C). A DFA can track the parity of the length while enforcing the a*b* shape for (A), and the value of $\#_a - \#_b \bmod 3$ for (C). (B) and (D) fail the pumping lemma.

Turing machines, decidability

Decidable / undecidable / RE problems, reductions. · 13 questions, 21 marks over 15 papers (≈1.4 marks/paper)

A. Past-paper questions (13)

2026 CS2 · Q.13 · 1 mark
Which one of the following statements is equivalent to the following assertion? Turing machine M decides the language L ⊆ {0,1}∗
(A) Turing machine M halts on all input strings in {0,1}∗
(B) Turing machine M accepts all input strings in L
(C) Turing machine M rejects all input strings in {0,1}∗ − L
(D) Turing machine M accepts all input strings in L and rejects all input strings in {0,1}∗ − L

2022 · Q.23 · 1 mark
Which of the following statements is/are TRUE?
(A) Every subset of a recursively enumerable language is recursive.
(B) If a language L and its complement L are both recursively enumerable, then L must be recursive.
(C) Complement of a context-free language must be recursive.
(D) If L1 and L2 are regular, then L1 L2 must be deterministic context-free.

2022 · Q.46 · 2 marks
Which of the following is/are undecidable?
(A) Given two Turing machines M1 and M 2 , decide if L(M1) L(M 2).
(B) Given a Turing machine M , decide if L (M) is regular.
(C) Given a Turing machine M , decide if M accepts all strings.
(D) Given a Turing machine M , decide if M takes more than 1073 steps on every string.

2021 · Q.12 · 1 mark [OCR]
Let (M) denote an encoding of an automaton M. Suppose that © = {0,1}. Which of the following languages is/are NOT recursive?
(A) | L = {(M)| M is a DFA such that L(M) = 0}
(B) | L = {(M)| Mis a DPA such that L(M) = 5°}
(C) | L={(M)| M isa PDA such that L(M) = 0}
(D) | L = {(M)| Misa PDA such that L(M) = 5°} CS, Set-1 - Copyright © GATE 2021

2021 · Q.39 · 2 marks [OCR]
For a Turing machine M, (M) denotes an encoding of M. Consider the following two languages. L; = {(M) | M takes more than 2021 steps on all inputs} Ly = {(M) | M takes more than 2021 steps on some input} Which one of the following options is correct?
(A) | Both LZ, and Ly are decidable.
(B) | ZL; is decidable and L» is undecidable.
(C) | L, is undecidable and Ly is decidable.
(D) | Both L, and Ly are undecidable. CS, Set-1 - Copyright © GATE 2021

2020 · Q.26 · 2 marks [OCR]
Which of the following languages are undecidable? Note that (M) indicates encoding of the Turing machine M. L, = {(M)| L(M) = 9} L2 = {(M,w,q)| M on input w reaches state q in exactly 100 steps} L3 = {(M)| L(M) is not recursive} L4 = {(M)| L(M) contains at least 21 members}
(A) L,L3, and L, only (8) L, and L; only (©) Lz and L3 only (0) Lz, L3, and L4 only

2019 · Q.34 · 2 marks [OCR]
Consider the following sets: Sl. Set of all recursively enumerable languages over the alphabet {0,1} S2. Set of all syntactically valid C programs S3. Set of all languages over the alphabet {0,1} S4. Set of all non-regular languages over the alphabet {0,1} Which of the above sets are uncountable?
(A) S1 and $2
(B) $3 and S4
(C) S2and $3
(D) S1 and $4

2018 · Q.7 · 1 mark
The set of all recursively enumerable languages is
(A) closed under complementation.
(B) closed under intersection.
(C) a subset of the set of all recursive languages.
(D) an uncountable set.

2018 · Q.36 · 2 marks
Consider the following problems. L(G) denotes the language generated by a grammar G. L(M) denotes the language accepted by a machine M. (I) For an unrestricted grammar G and a string w, whether w ∈ L(G) (II) Given a Turing machine M, whether L(M) is regular (III) Given two grammars G1 and G2 , whether L(G1) = L(G2) (IV) Given an NFA N, whether there is a deterministic PDA P such that N and P accept the same language. Which one of the following statements is correct?
(A) Only I and II are undecidable
(B) Only III is undecidable
(C) Only II and IV are undecidable
(D) Only I, II and III are undecidable CS 11/20 GATE 2018

2017 CS1 · Q.39 · 2 marks [OCR]
Let A and B be finite alphabets and let # be a symbol outside both 4 and B. Let f be a total . * a . . . . . function from 4 toB . We say fis computable if there exists a Turing machine M which given an : : hl . a * input x in A . always halts with f/x) on its tape. Let L f denote the language {x# f(x)|xeA }. Which of the following statements is true:
(A) fis computable if and only if L 73s recursive.
(B) fis computable if and only if Z f is recursively enumerable.
(C) Iffis computable then L f is recursive. but not conversely.
(D) Iffis computable then L f is recursively enumerable. but not conversely. =====PAGE 18=====

2017 CS2 · Q.41 · 2 marks [OCR]
Let L(R) be the language represented by regular expression R. Let L(G) be the language generated by a context free grammar G. Let L(M) be the language accepted by a Turing machine M. Which of the following decision problems are undecidable? I. Given a regular expression R and a string w. is w € L(R)? II. Given a context-free grammar G. is L(G) = 0? III. Given a context-free grammar G. is L(G) = * for some alphabet = ? IV. Given a Turing machine M and a string w. is w € L(M)?
(A) Land IV only
(B) II and II only
(C) I, Wland IV only
(D) I and IV only

2016 · Q.17 · 1 mark
Which of the following decision problems are undecidable? I. Given NFAs N1 and N2 , is L(N1) ∩ L(N2) = Φ? II. Given a CFG G = (N, Σ, P, S) and a string x ∈ Σ∗ , does x ∈ L(G)? III. Given CFGs G1 and G2 , is L(G1) = L(G2)? IV. Given a TM M, is L(M) = Φ?
(A) I and IV only
(B) II and III only
(C) III and IV only
(D) II and IV only

2016 · Q.44 · 2 marks
Let X be a recursive language and Y be a recursively enumerable but not recursive language. Let W and Z be two languages such that Y reduces to W , and Z reduces to X (reduction means the standard many-one reduction). Which one of the following statements is TRUE?
(A) W can be recursively enumerable and Z is recursive.
(B) W can be recursive and Z is recursively enumerable.
(C) W is not recursively enumerable and Z is recursive.
(D) W is not recursively enumerable and Z is not recursive.

B. New practice questions with solutions

1. (1 mark · MCQ) Which of the following problems is UNDECIDABLE?

A. Given a DFA M, is L(M) empty?
B. Given a CFG G, is L(G) empty?
C. Given a CFG G over Σ, is L(G) = Σ*?
D. Given a regular expression r, is L(r) infinite?

(C). Universality of a CFG is undecidable. Emptiness of a DFA or CFG and finiteness of a regular language are decidable.

2. (2 marks · MSQ) Which statements are TRUE?

A. The complement of a recursively enumerable language that is not recursive is not recursively enumerable.
B. If L and its complement are both recursively enumerable, then L is recursive.
C. The set of Turing machines that halt on the empty tape is recursive.
D. Every subset of a recursive language is recursive.

(A), (B). (A) follows from (B). (C) is the halting problem on the empty tape, which is undecidable. (D) is false because every language over Σ is a subset of the recursive language Σ^*.